

BOW VALLEY, AB, Canada

WMO: 712320

Lat: 51.083N

Lon: 115.066W

Elev: 4257

StdP: 12.57

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-23.2	-15.1	-29.5	1.2	-22.8	-21.4	2.0	-14.4	24.7	29.1	22.8	30.1	1.0	50	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.7	83.9	58.2	80.2	57.2	76.4	55.9	61.8	75.7	59.8	74.0	57.9	72.1	7.6	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.0	81.3	66.5	54.5	74.1	63.5	52.4	68.6	61.3	29.7	75.9	28.1	74.2	26.8	72.1	69.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 19.8	17.4	15.3	DB	-34.3	89.8	6.8	3.6	-39.2	92.4	-43.1	94.5	-46.9	96.5	-51.9	99.1	
(5)			WB	-34.5	65.5	6.7	2.5	-39.3	67.3	-43.3	68.8	-47.1	70.2	-51.9	72.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	39.0	20.8	22.1	28.7	37.7	46.3	53.4	60.0	58.2	50.6	40.4	28.7	20.2	(6)
(7)		DBStd	18.00	16.40	14.95	14.20	9.05	7.97	5.99	5.73	6.16	8.26	9.27	13.21	15.01	(7)
(8)		HDD50	4908	906	782	660	377	166	31	4	10	93	317	640	922	(8)
(9)		HDD65	9514	1370	1202	1125	819	579	351	174	222	433	764	1088	1387	(9)
(10)		CDD50	901	0	0	1	8	52	133	313	264	111	18	1	0	(10)
(11)		CDD65	33	0	0	0	0	0	3	17	12	1	0	0	0	(11)
(12)		CDH74	1313	0	0	0	0	26	99	584	481	123	0	0	0	(12)
(13)		CDH80	332	0	0	0	0	1	21	169	121	20	0	0	0	(13)
(14)	Wind	WSAvg	6.5	8.5	7.3	6.6	5.8	5.5	5.5	5.4	5.0	5.6	6.8	7.6	8.6	(14)
(15)	Precipitation	PrecAvg	22.2	0.7	0.7	1.3	1.7	3.1	4.3	2.3	2.4	2.4	1.5	1.2	0.7	(15)
(16)		PrecMax	31.1	1.8	1.3	3.0	4.9	5.7	14.1	7.4	6.0	5.8	3.0	3.3	2.0	(16)
(17)		PrecMin	14.5	0.0	0.0	0.2	0.3	0.6	1.1	0.6	0.0	0.2	0.5	0.4	0.2	(17)
(18)		PrecStd	4.6	0.4	0.4	0.6	1.1	1.3	2.8	1.5	1.3	1.5	0.6	0.7	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.7	52.0	59.6	68.7	77.3	83.4	89.7	87.8	83.1	68.4	57.8	50.3	(19)
(20)			MCWB	40.8	38.6	41.9	47.7	52.5	57.5	59.9	58.3	56.1	50.1	43.5	39.2	(20)
(21)		2%	DB	46.8	47.6	53.4	62.8	72.5	76.3	84.6	83.4	77.5	62.5	52.6	45.3	(21)
(22)			MCWB	36.5	36.4	39.6	45.2	50.6	55.4	59.7	57.3	54.4	47.0	41.3	35.8	(22)
(23)		5%	DB	43.4	44.0	49.5	57.7	67.8	72.0	80.6	79.6	72.8	58.5	48.6	41.7	(23)
(24)			MCWB	34.3	34.0	37.7	42.4	49.1	53.8	58.3	57.0	52.9	45.2	39.0	33.4	(24)
(25)		10%	DB	40.2	40.0	45.9	53.2	63.1	67.9	76.3	75.4	67.1	54.5	44.6	38.2	(25)
(26)			MCWB	32.4	31.4	35.5	40.1	47.3	51.7	57.2	56.1	50.7	42.9	36.1	31.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.6	39.7	43.2	48.7	55.3	61.6	66.3	63.4	58.8	51.6	44.8	40.0	(27)
(28)			MCDWB	50.0	50.3	55.9	65.8	71.1	74.1	79.2	76.5	76.2	65.9	54.6	49.2	(28)
(29)		2%	WB	37.3	36.9	40.7	46.0	52.5	58.0	62.8	61.0	56.0	48.5	42.0	36.2	(29)
(30)			MCDWB	45.5	46.5	51.8	60.7	67.1	71.4	76.7	74.7	72.9	60.5	51.4	44.2	(30)
(31)		5%	WB	34.8	34.2	38.2	43.3	50.3	55.4	60.7	58.9	53.8	45.9	39.5	33.7	(31)
(32)			MCDWB	42.3	43.0	48.3	56.1	64.6	68.5	74.9	74.4	69.6	56.9	48.1	41.2	(32)
(33)		10%	WB	32.6	31.8	35.9	40.7	48.3	53.3	58.8	57.1	51.5	43.5	36.5	31.2	(33)
(34)			MCDWB	39.9	39.4	45.5	51.7	62.1	64.7	72.4	71.4	65.2	53.4	44.1	37.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.4	20.3	20.9	21.4	23.3	22.8	27.7	28.2	24.8	19.2	15.5	15.4	(35)
(36)			MCDBR	15.2	19.5	22.2	29.6	33.1	32.7	37.5	37.7	36.2	25.4	18.0	14.6	(36)
(37)			MCWBR	9.8	12.0	12.6	16.0	16.3	15.6	16.2	16.3	16.6	13.8	11.1	10.0	(37)
(38)		5% WB	MCDBR	14.9	19.8	21.9	27.7	29.7	29.0	32.2	33.2	33.8	24.7	19.1	15.2	(38)
(39)			MCWBR	10.1	12.8	13.2	15.5	15.4	15.0	16.1	16.7	16.6	14.0	12.3	11.1	(39)
(40)		Clear-Sky Solar Irradiance	taub	0.251	0.265	0.286	0.337	0.346	0.359	0.348	0.362	0.320	0.282	0.257	0.240	(40)
(41)	taud		2.437	2.427	2.418	2.291	2.362	2.366	2.409	2.383	2.482	2.574	2.516	2.426	(41)	
(42)	Ebn at Noon		256	281	293	286	288	284	285	275	278	272	252	242	(42)	
(43)	Edh at Noon		18	24	29	37	36	37	35	34	27	21	16	15	(43)	
(44)	All-Sky Solar Radiation	RadAvg	354	656	1037	1443	1679	1773	1991	1652	1164	723	384	279	(44)	
(45)		RadStd	23	44	69	101	141	146	143	117	133	70	25	17	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page